CENSOR: Defense Against Gradient Inversion via Orthogonal Subspace Bayesian Sampling

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Introduction

Federated Learning is a distributed machine learning approach where multiple users collaboratively train a global model without sharing their raw local data. Instead of transferring datasets directly, only gradients or model updates are shared with a central server.

Although Federated Learning improves privacy, recent studies have shown that attackers can reconstruct sensitive training samples from shared gradients using Gradient Inversion Attacks.

To address this issue, the CENSOR framework introduces a privacy-preserving defense mechanism using Orthogonal Subspace Decomposition and Bayesian Sampling to protect gradients while maintaining model accuracy.

Problem Statement

In federated learning systems, gradients shared between clients and servers may unintentionally leak sensitive information about user data.

Attackers can exploit this vulnerability using Gradient Inversion Attacks, where private images or training samples are reconstructed from shared gradients.

Existing defense techniques such as Differential Privacy and noise injection often reduce model accuracy or negatively affect convergence.

Therefore, there is a need for a defense mechanism that provides strong privacy protection while preserving training utility and model performance.

Project Objectives

- Study Gradient Inversion Attacks in Federated Learning
- Analyze privacy leakage from shared gradients
- Understand Orthogonal Subspace Projection
- Explore Bayesian Sampling techniques
- Prevent reconstruction of private training data
- Preserve model accuracy during training

Background

Federated Learning enables decentralized AI training where users keep their data locally while sharing gradients with a central server.

However, gradients can reveal important information about training samples. Recent attacks demonstrate that private images and text data can be reconstructed from gradients alone.

Traditional defenses rely heavily on noise addition, which can degrade model performance. CENSOR improves privacy protection by combining orthogonal decomposition with probabilistic Bayesian Sampling.

System Architecture

Client Data $\rightarrow$ Gradient Computation

Gradient $\rightarrow$ Orthogonal Subspace Decomposition

Bayesian Sampling $\rightarrow$ Protected Gradient

Protected Gradient $\rightarrow$ Central Server

The server uses protected gradients for global model training while reducing privacy leakage risks.

Orthogonal Subspace Decomposition

CENSOR separates gradients into orthogonal components to isolate sensitive information.

Mathematically:

$$g = g_{\parallel} + g_{\perp}$$

Where:

- $g_{\parallel}$ represents useful learning information
- $g_{\perp}$ contains privacy-sensitive directions

Sensitive gradient components are suppressed or perturbed before sharing.

Bayesian Sampling

Instead of sharing exact gradients, CENSOR generates protected gradients using Bayesian Sampling.

Bayesian Sampling introduces uncertainty by sampling multiple plausible gradients from a posterior distribution.

This makes it difficult for attackers to determine the exact original training data while still preserving learning utility.

Advantages:

- Increased privacy
- Reduced reconstruction accuracy
- Better training performance

Methodology

The CENSOR framework follows these steps:

1. Compute local gradients from client data
2. Perform orthogonal subspace decomposition
3. Identify privacy-sensitive directions
4. Apply Bayesian Sampling
5. Generate protected gradients
6. Send protected gradients to server
7. Update global model

Advantages of CENSOR

- Strong defense against Gradient Inversion Attacks
- Preserves model utility and convergence
- Introduces probabilistic uncertainty
- Reduces privacy leakage
- More effective than excessive noise injection
- Suitable for secure Federated Learning systems

Limitations

Although CENSOR improves privacy protection, some challenges still exist:

- Additional computational overhead
- Increased training complexity
- Parameter tuning required
- Adaptive attackers may still partially recover information

Further research is needed to improve scalability and robustness against advanced attacks.

Results and Analysis

The CENSOR framework significantly reduces the success rate of Gradient Inversion Attacks.

Observations:

- Reconstructed images become unclear or distorted
- Privacy leakage decreases substantially
- Model accuracy remains relatively high
- Training stability is preserved

This demonstrates that CENSOR achieves a balance between privacy and learning performance.

Conclusion

CENSOR is an effective privacy-preserving defense framework for Federated Learning systems.

By combining Orthogonal Subspace Decomposition and Bayesian Sampling, the framework protects sensitive gradients against inversion attacks while preserving training accuracy.

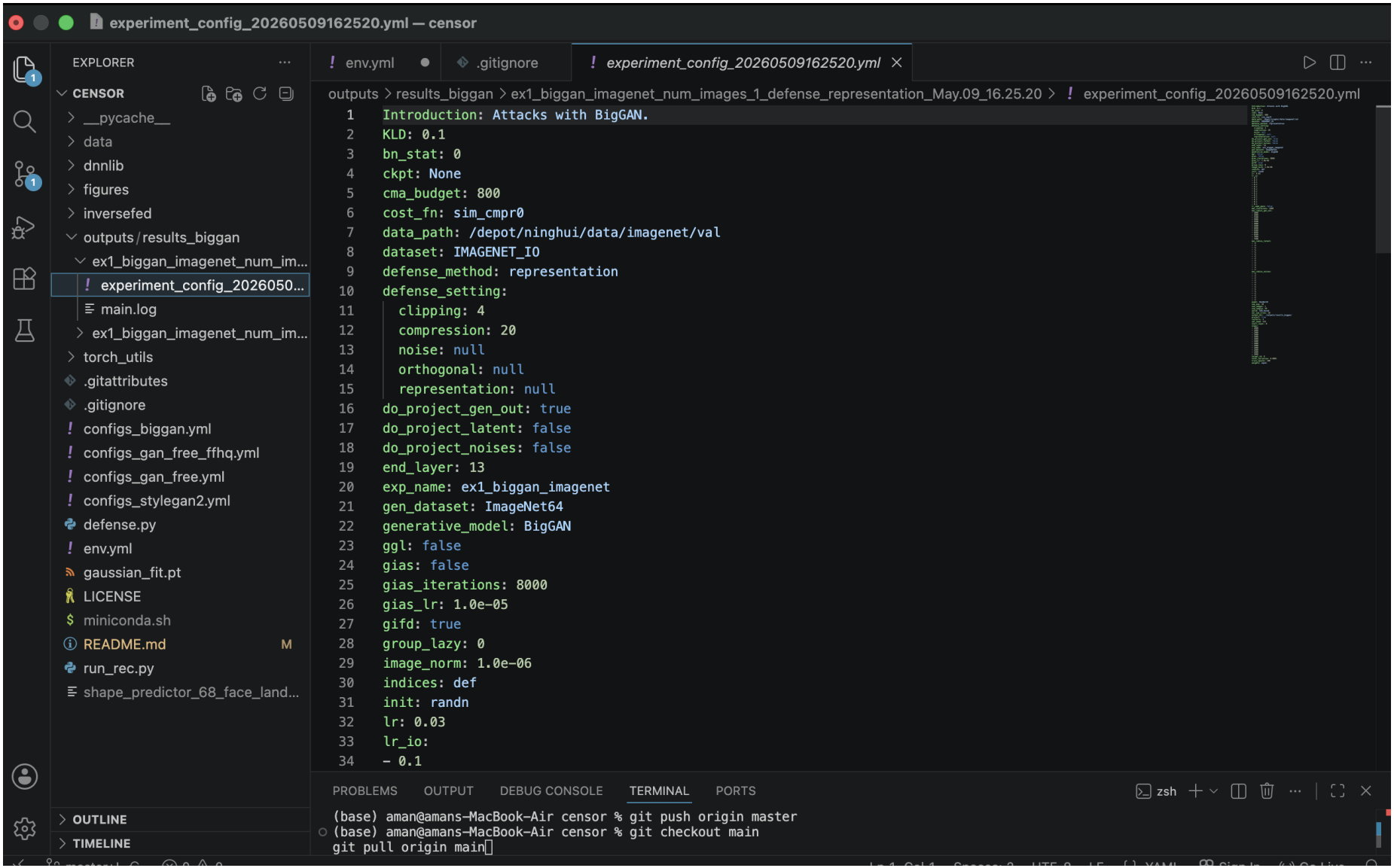
The approach demonstrates how probabilistic gradient protection techniques can improve secure distributed machine learning systems.

Future work can focus on reducing computational overhead and improving resistance against adaptive adversarial attacks.

Output

DA	Defense	IG [5]				GI [9]				GGL [8]				GIAS [6]				GIFD [4]			
		MSE↑	LPIPS↑	PSNR↓	SSIM↓	MSE↑	LPIPS↑	PSNR↓	SSIM↓	MSE↑	LPIPS↑	PSNR↓	SSIM↓	MSE↑	LPIPS↑	PSNR↓	SSIM↓	MSE↑	LPIPS↑	PSNR↓	SSIM↓
ImageNet	No Defense	0.0195	0.5574	17.819	0.2309	0.0191	0.5402	17.908	0.2400	0.0453	0.5952	13.873	0.0745	0.0191	0.4795	18.452	0.3099	0.0130	0.3782	21.364	0.4528
	Noise [13]	0.0246	0.6294	16.338	0.1754	0.0269	0.6300	15.883	0.1549	0.0410	0.5697	14.252	0.0817	0.0253	0.5947	16.601	0.1854	0.0196	0.5380	18.166	0.2686
	Clipping [14]	0.0167	0.5008	18.883	0.3128	0.0383	0.7302	14.844	0.0106	**0.0477**	0.5823	**13.520**	0.0749	0.0203	0.4825	18.738	0.3186	0.0150	0.4433	19.547	0.3798
	Sparsi [15]	0.0137	0.4945	19.383	0.3419	0.0157	0.4941	18.799	0.3099	0.0456	0.6080	13.743	0.0776	0.0135	0.3981	20.483	0.4182	0.0179	0.4444	19.486	0.3686
	Soteria [16]	**0.0662**	**0.7596**	**12.220**	0.0135	**0.0682**	0.7485	**12.215**	0.0134	0.0461	0.5986	13.879	0.0708	0.0245	0.4986	17.646	0.2664	0.0139	0.3967	20.602	0.4335
	CENSOR	0.0600	0.7551	12.463	**0.0067**	0.0416	**0.8615**	14.446	**0.0021**	0.0419	**0.7912**	14.262	**0.0094**	**0.0650**	**0.7591**	**12.266**	**0.0139**	**0.0507**	**0.7610**	**13.323**	**0.0094**
FFHQ	No Defense	0.0143	0.5247	18.666	0.4209	0.0194	0.5692	17.246	0.3490	0.0421	0.5424	14.167	0.1953	0.0173	0.4228	18.738	0.4792	0.0149	0.4353	20.116	0.5102
	Noise [13]	0.0311	0.6666	15.298	0.2377	0.0388	0.7131	14.243	0.1507	0.0454	0.5784	13.866	0.1752	0.0225	0.5878	17.070	0.3449	0.0190	0.5250	17.953	0.3972
	Clipping [14]	0.0262	0.6245	15.963	0.2865	0.0593	0.8223	12.514	0.0080	0.0381	0.4881	14.571	0.2286	0.0171	0.4267	18.693	0.4808	0.0147	0.4353	20.113	0.5139
	Sparsi [15]	0.0225	0.5745	17.113	0.3411	0.0278	0.6189	16.068	0.2889	0.0421	0.5479	14.182	0.2035	0.0146	0.4057	19.490	0.5130	0.0125	0.4070	20.581	0.5423
	Soteria [16]	**0.1124**	**0.8446**	**9.662**	0.0165	**0.1105**	**0.8467**	**9.753**	0.0157	0.0438	0.5279	13.842	0.2014	0.0114	0.3560	20.551	0.5489	0.0165	0.4686	19.296	0.4789
	CENSOR	0.1009	0.8347	10.125	**0.0080**	0.0992	0.8335	10.550	**0.0061**	**0.0823**	**0.8155**	**10.947**	**0.0219**	**0.1108**	**0.7895**	**9.613**	**0.0268**	**0.1037**	**0.8097**	**9.904**	**0.0195**
CIFAR-10	No Defense	0.0023	0.0905	26.324	0.8139	0.0192	0.3909	17.170	0.4574	0.0275	0.5569	15.601	0.1099	0.0009	0.0333	30.414	0.9276	0.0201	0.5297	16.968	0.2408
	Noise [13]	0.0052	0.2229	22.810	0.7010	0.0060	0.2587	22.242	0.7241	0.0292	0.5974	15.339	0.1002	0.0022	0.2600	26.660	0.6555	0.0137	0.5166	18.648	0.3581
	Clipping [14]	0.0036	0.1050	24.396	0.8048	0.0496	0.5317	13.042	0.0482	**0.0689**	0.6125	**11.621**	0.0935	0.0014	0.0568	28.567	0.8863	0.0228	0.5492	16.426	0.2179
	Sparsi [15]	0.0093	0.3210	20.311	0.5933	0.0041	0.1631	23.841	0.7614	0.0583	0.6491	12.340	0.1407	0.0003	0.0100	35.652	0.9631	0.0297	0.4557	15.267	0.2432
	Soteria [16]	0.0558	0.5846	12.537	0.1222	**0.0891**	0.6986	**10.501**	0.0378	0.0326	0.6274	14.867	0.0731	0.0012	0.0461	29.308	0.9050	0.0239	0.5577	16.222	0.3907
	CENSOR	**0.0939**	**0.6884**	**10.272**	**0.0112**	0.0806	**0.7006**	10.937	**0.0167**	0.0260	**0.7159**	15.857	**0.0179**	**0.0789**	**0.6819**	**11.027**	**0.0789**	**0.0916**	**0.6398**	**10.381**	**0.1452**

Our outputs-1



Our outputs-2

experiment_config_20260509162520.yml — censor

EXPLORER

CENSOR

- __pycache__
- data
- dnnlib
- figures
- inversefed
- outputs / results_biggan
  - ex1_biggan_imagenet_num_im...
    - experiment_config_2026050...
    - main.log
  - ex1_biggan_imagenet_num_im...
- torch_utils
- .gitattributes
- .gitignore
- configs_biggan.yml
- configs_gan_free_ffhq.yml
- configs_gan_free.yml
- configs_stylegan2.yml
- defense.py
- env.yml
- gaussian_fit.pt
- LICENSE
- miniconda.sh
- README.md M
- run_rec.py
- shape_predictor_68_face_land...

OUTLINE

TIMELINE

env.yml | .gitignore | experiment_config_20260509162520.yml

outputs > results_biggan > ex1_biggan_imagenet_num_images_1_defense_representation_May.09_16.25.20 > experiment_config_20260509162520.yml

```yaml
lr: 0.03
lr_io:
- 0.1
- 0.1
- 0.1
- 0.1
- 0.1
- 0.1
- 0.1
- 0.1
- 0.1
- 0.1
- 0.1
- 0.1
- 0.1
lr_same_pace: false
max_iterations: 1000
max_radius_gen_out:
- 1000
- 2000
- 2500
- 3000
- 3500
- 4000
- 4500
- 5000
- 5500
- 6000
- 6500
- 7000
- 7500
max_radius_latent:
- -1
- -1
- -1
- -1
- -1
- -1
```

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Our outputs-3

experiment_config_20260509162520.yml — censor

EXPLORER

CENSOR
- __pycache__
- data
- dnnlib
- figures
- inversefed
- outputs / results_biggan
  - ex1_biggan_imagenet_num_im...
    - experiment_config_2026050...
    - main.log
  - ex1_biggan_imagenet_num_im...
- torch_utils
- .gitattributes
- .gitignore
- configs_biggan.yml
- configs_gan_free_ffhq.yml
- configs_gan_free.yml
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- defense.py
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- README.md M
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- shape_predictor_68_face_land...

OUTLINE

TIMELINE

env.yml | .gitignore | experiment_config_20260509162520.yml

outputs > results_biggan > ex1_biggan_imagenet_num_images_1_defense_representation_May.09_16.25.20 > experiment_config_20260509162520.yml

```yaml
  - -1
  - -1
  - -1
  - -1
  - -1
  - -1
model: ResNet18
num_exp: 10
num_images: 1
num_sample: 50
optim: GAN_based
our_num_tries: 20
output_dir: ./outputs/results_biggan/
project: true
restarts: 4
set_seed: 123
start_layer: 0
steps:
- 1000
- 1000
- 1000
- 1000
- 1000
- 1000
- 1000
- 1000
- 1000
- 1000
- 1000
- 1000
- 1000
target_id: 0
total_variation: 0.0001
train_epochs: 100
weights: equal
```

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Thank You